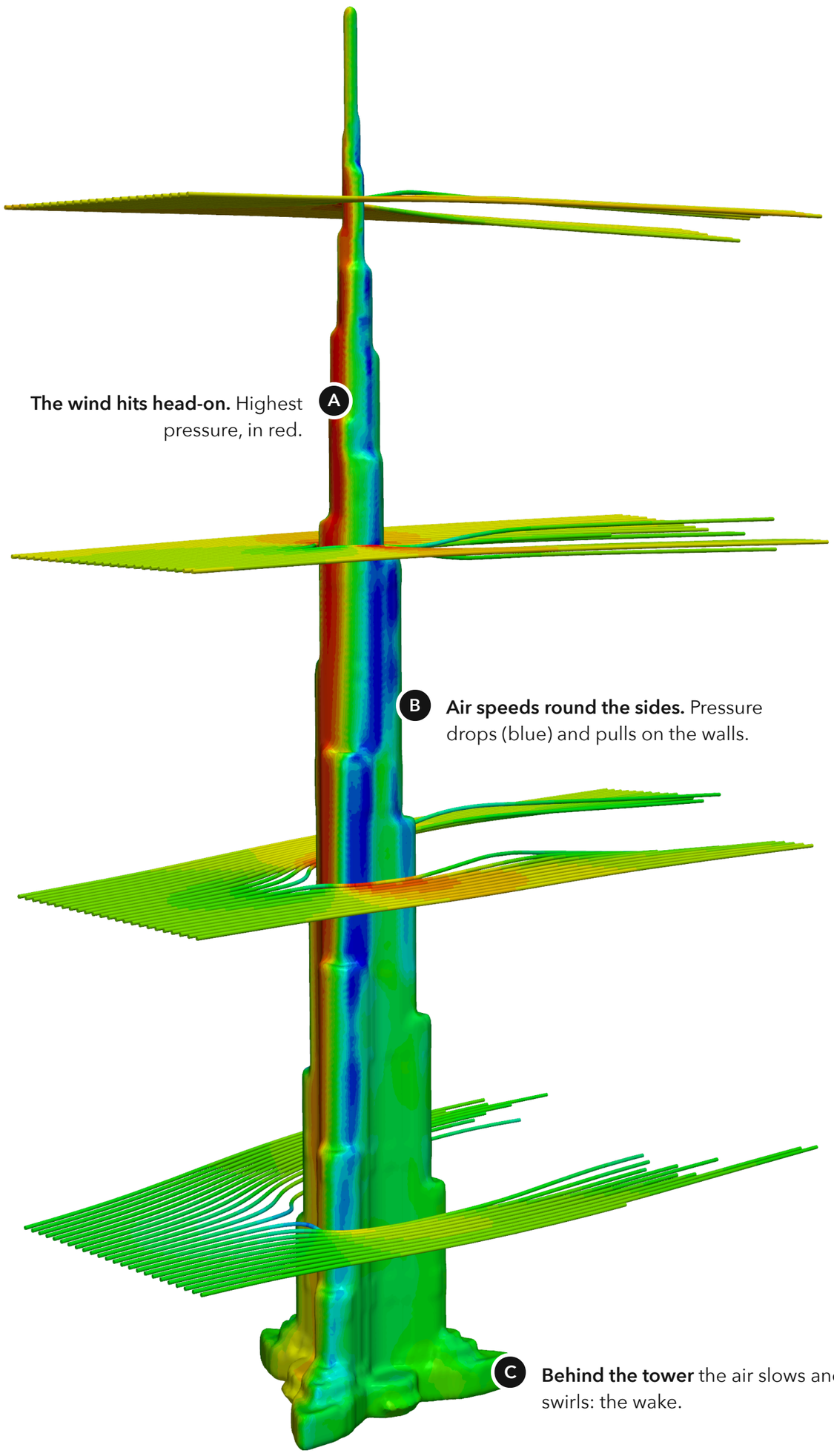
828 metres, standing in the wind.

We solved the wind around the Burj Khalifa, the tallest building on Earth, from nine directions with the Navier–Stokes equations, to see where it pushes hardest and why.



The wind hits head-on. Highest pressure, in red.

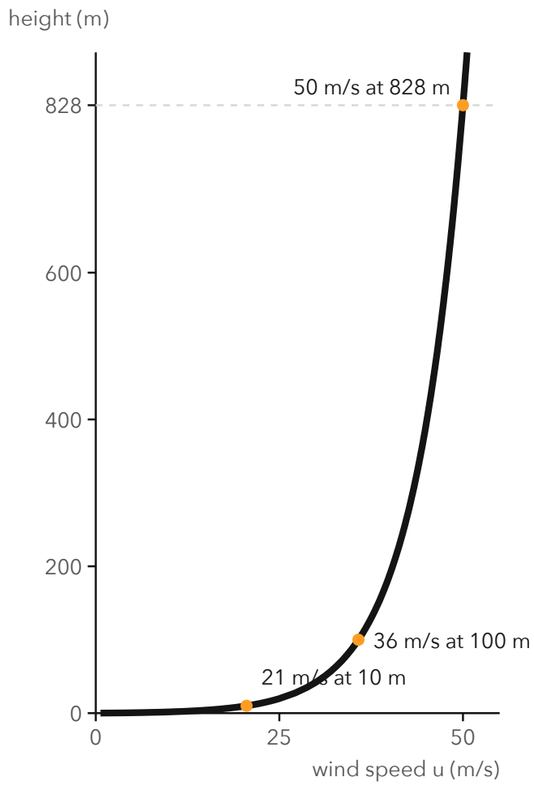
Air speeds round the sides. Pressure drops (blue) and pulls on the walls.

Behind the tower the air slows and swirls: the wake.

1 The ground slows the wind.

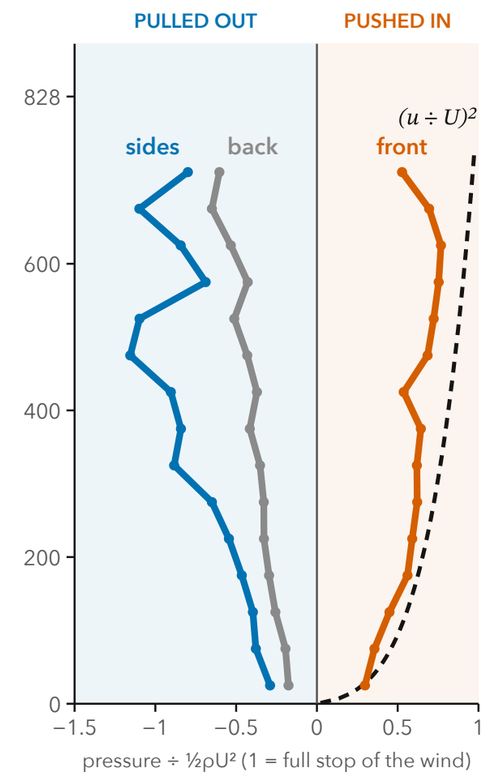
$u(z) \propto \ln((z + z_0) \div z_0)$

a logarithm: fast high up, slow near the ground.

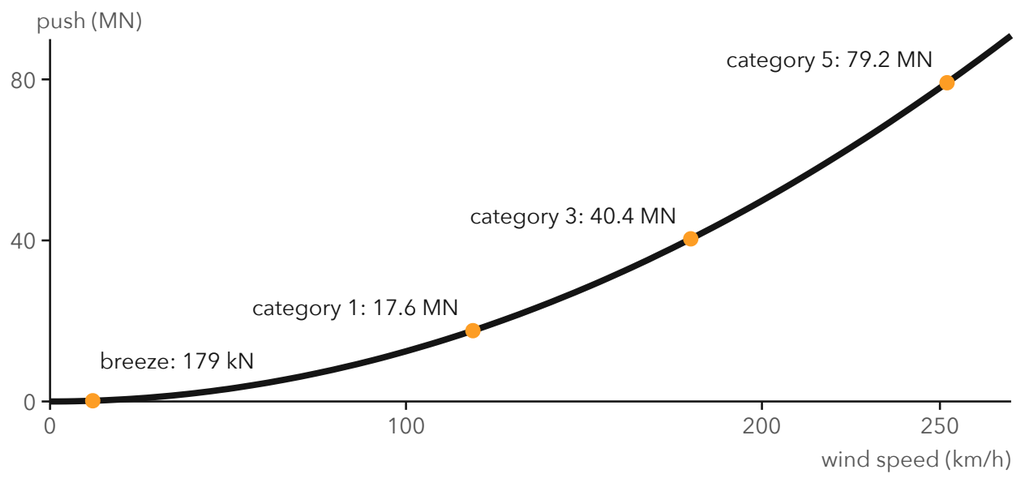


2 So the push grows with height.

The front is pushed in, harder higher up, growing like $(u \div U)^2$ (dashed). The sides and back are sucked outward.



3 $F \propto v^2$: double the wind, four times the push.



A category 5 hurricane pushes with 79.2 MN, about the weight of 8,000 tonnes. Turning effect at the base = force × distance: 23.7 GN·m ÷ 79.2 MN = 299 m up the tower.

Surface pressure

Low: suction — High: wind hits

Air speed

Still — 234 km/h

The same equation, a different job.

$$\rho(\partial v/\partial t + (v\cdot\nabla)v) = -\nabla p + \mu\nabla^2 v$$

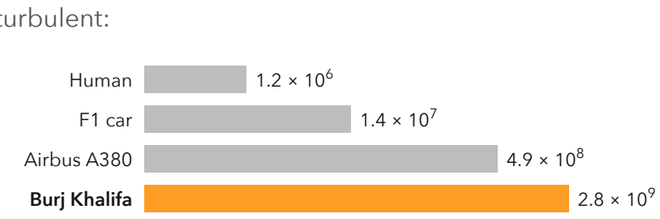
A steady wind, so the time term drops out. Size of each term near the tower:



m/s² per kilogram of air. Pressure and inertia balance; friction is ten times smaller.

How a building is different.

- **It stands on the ground.** A plane flies in even air; a tower meets wind that changes with height.
- **It is blunt.** Air breaks away at its edges, leaving a big wake. It needs no lift, only to resist the push.
- **It is huge.** Reynolds number $Re = VD \div \nu$, so the flow is fully turbulent:



Turn the wind: the push barely changes.

< 6 %



Category 5 push from all nine directions: 74.9 to 79.2 MN. The Y-shaped floor plan spreads it evenly.

SDG 11

Sustainable cities and communities

Make cities inclusive, safe, resilient and sustainable. Knowing the wind's push lets engineers build towers that stay safe in storms without more steel and concrete than they need.

Solved with OpenFOAM, k-ω SST · 9 wind directions, about 216,000 cells each · 180 km/h at the roof, ground roughness 0.5 m

NavFlow · NPS International School STEM Exhibition